# Identifying Deployment Environments Using RSSI Fluctuations and Deep Learning

Project Report

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## Abstract

This report presents an end to end study on classifying deployment environments from Received Signal Strength Indicator (RSSI) signals collected by low-power IoT nodes. The project uses five environments bridge, forest, garden, lake, and river and preprocesses RSSI signals through differentiation, min-max normalization, and sliding window segmentation before applying Convolutional Neural Network (CNN) and Residual Network (ResNet) models. Exploratory visual analysis shows that the environments have overlapping absolute RSSI ranges but different fluctuation behavior. On the same-dataset evaluation, the CNN model reached 82% accuracy and the ResNet model reached 94% accuracy. The ResNet confusion matrix shows very strong performance for bridge, forest, lake, and river, while garden remained the most difficult class. Feature-importance analysis further suggests that discriminative information is distributed across the full RSSI window rather than concentrated at a single point.


## 1. Dataset Overview

The uploaded dataset contains 835,971 RSSI records with nine fields: environment, test, sensor, sample_index, timestamp, address, RSSI, LQI, and source_file. Data was collected from two receiving sensor nodes while a third node transmitted packets. The available environments are bridge, forest, garden, lake, and river. This satisfies the five-environment requirement

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 835971 entries, 0 to 835970
Data columns (total 9 columns):
 #   Column        Non-Null Count   Dtype
---  ------        --------------   -----
 0   environment   835971 non-null  object
 1   test          835971 non-null  int64
 2   sensor        835971 non-null  object
 3   sample_index  835971 non-null  int64
 4   timestamp     835971 non-null  object
 5   addr          835971 non-null  object
 6   rssi          835971 non-null  int64
 7   lqi           835971 non-null  int64
 8   source_file   835971 non-null  object
dtypes: int64(4), object(5)
memory usage: 57.4+ MB
```

*Figure 1. Preview of the raw dataset showing environment, test number, sensor ID, timestamp, RSSI, LQI, and source file.*

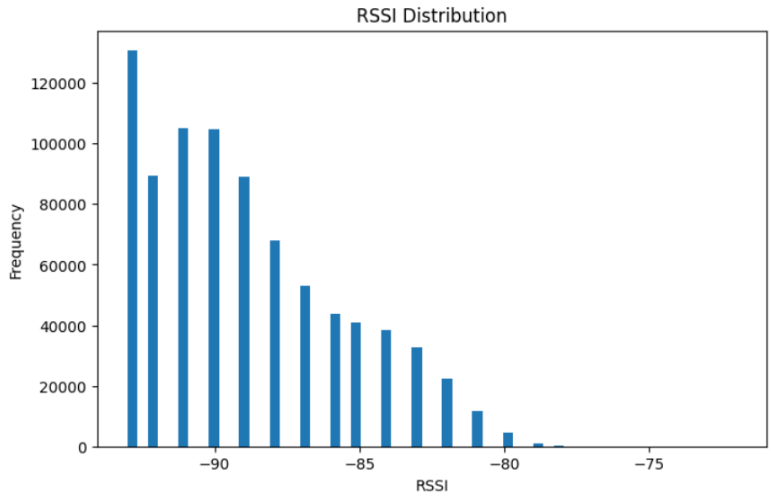



*Figure 2. Dataset structure summary. The dataset contains 835,971 non-null rows and nine columns.*

The imbalance visible in the raw counts matters for interpretation. A model can perform very well overall while still struggling on the least represented environment. This pattern appears later in the class-wise metrics, where garden is consistently the hardest class.

## 2. Exploratory Data Analysis

The first stage of analysis examined whether absolute RSSI alone already separates the environments. The overall RSSI histogram is concentrated roughly between −93 dBm and −80 dBm, with a strong peak around the low −90s. This indicates that the signal ranges of the five environments overlap heavily. Therefore, a good classifier must learn temporal fluctuation patterns rather than depend only on average signal level.

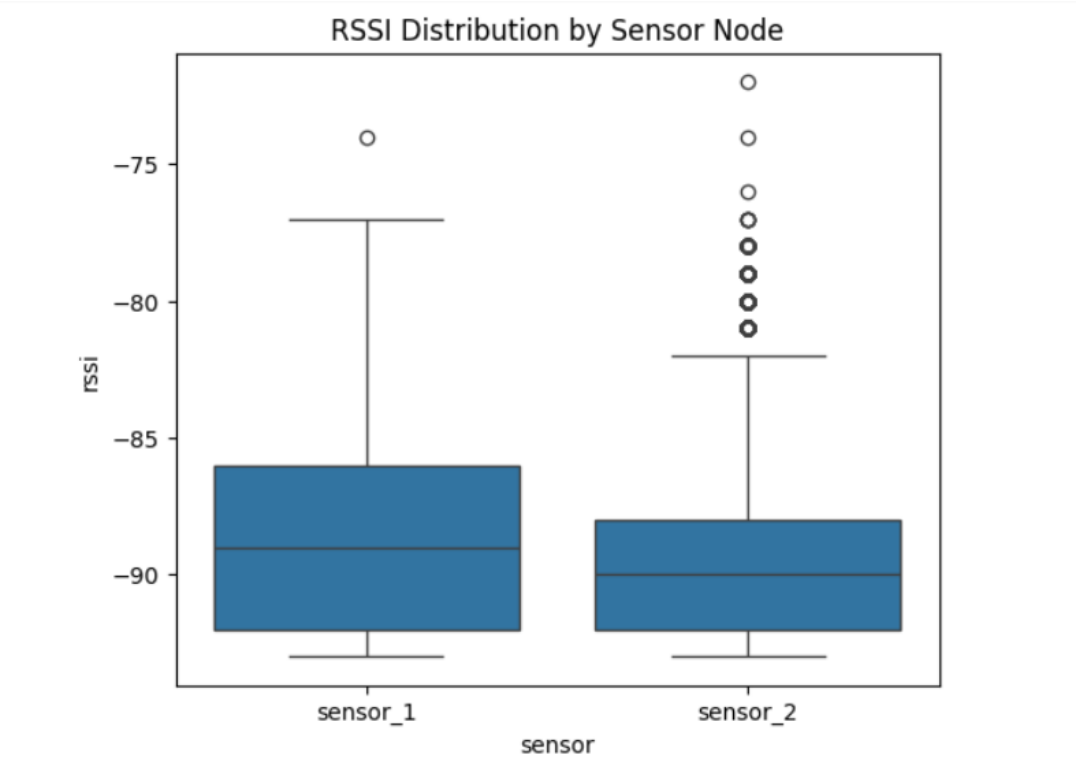



*Figure 3. Overall RSSI distribution. Most observations cluster in the low −90 dBm range, indicating substantial overlap in absolute signal strength.*

The boxplot by sensor node shows that the two receiving nodes have very similar medians, although sensor_2 exhibits more high-value outliers. This suggests that device-specific characteristics exist, but the two sensors are still similar enough that environment classification remains a non-trivial task.

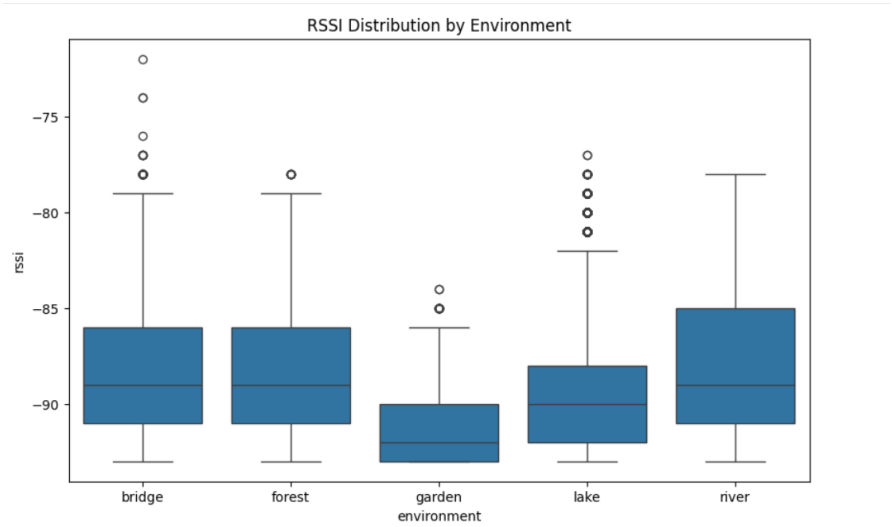



*Figure 4. RSSI distribution by receiving sensor. The medians are close, while sensor_2 shows more outliers.*

The environment boxplot provides an important insight: garden has the lowest median RSSI, while bridge, forest, and river overlap strongly. Lake also overlaps with bridge and river. This means that a model using only static RSSI

statistics would likely confuse several classes. The later success of CNN and ResNet therefore comes from learning dynamic variation patterns across time rather than simple level differences.

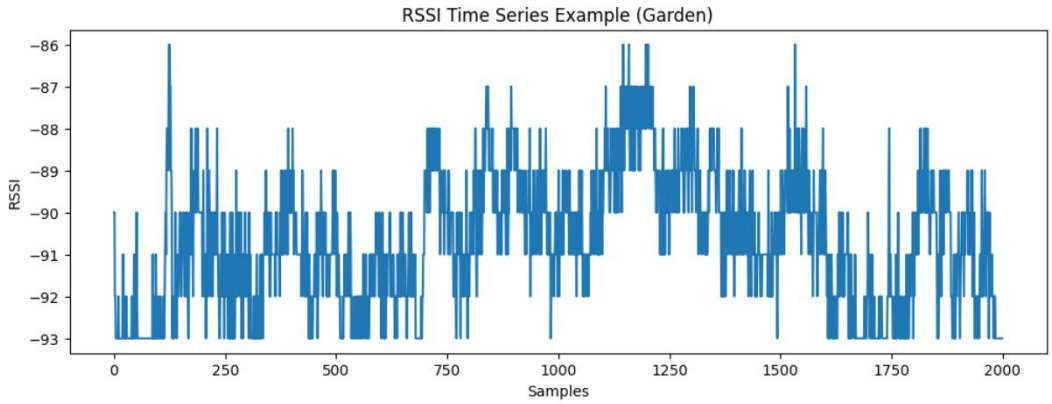



*Figure 5. RSSI distribution by environment. Garden is shifted lower, but bridge, forest, lake, and river still overlap considerably.*

The garden time-series example confirms that RSSI is not stationary. Even within a single environment, the signal oscillates over time, which is exactly the type of structure that one-dimensional deep models can learn.

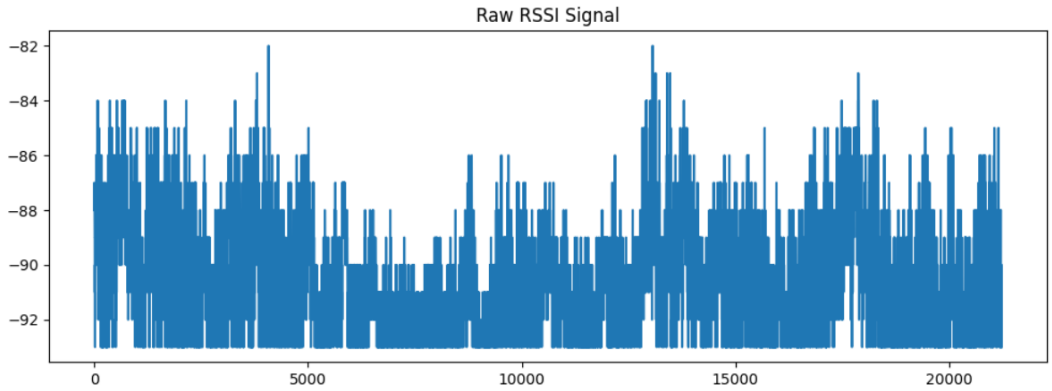



*Figure 6. Example RSSI time series from the garden environment, illustrating short-term fluctuations around a slowly varying trend.*

## 3. Preprocessing Pipeline

The assignment explicitly requires three preprocessing operations before model training: differentiation of the raw signal, normalization to the 0–1 range, and segmentation into overlapping frames. The submitted preprocessing follows this order. Differentiation suppresses slowly changing offsets and focuses the model on local changes in link quality. Normalization removes scale differences and makes the inputs stable for training. Sliding windows then convert the long sequence into fixed-length samples suitable for CNN and ResNet architectures.

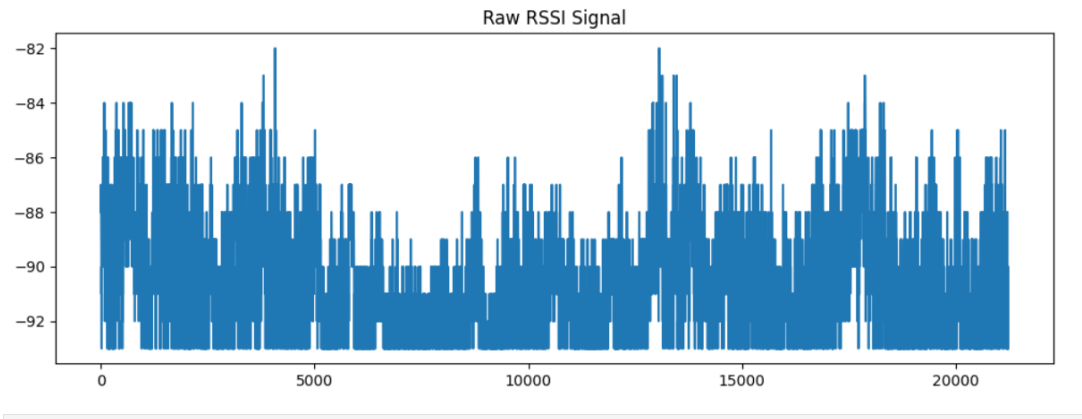



*Figure 7. Raw RSSI signal over one long run. The sequence contains both slow trends and short fluctuations.*

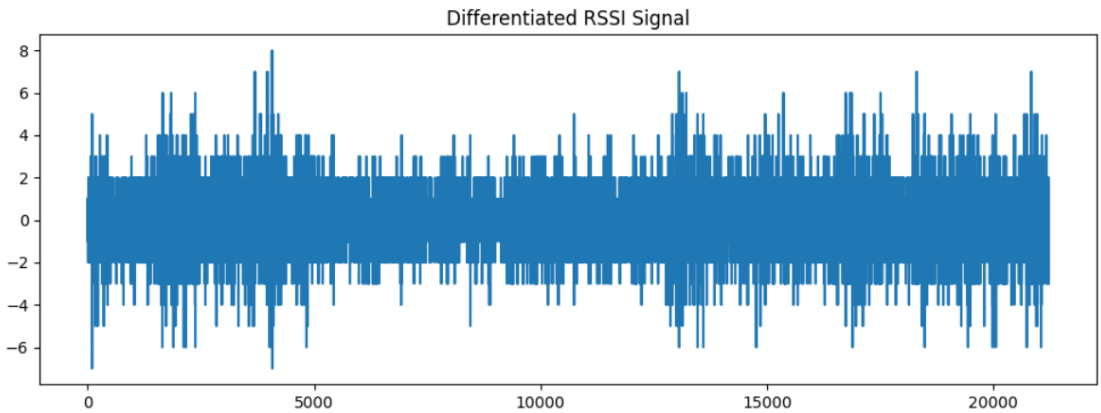



*Figure 8. Differentiated RSSI signal. The transformed sequence emphasizes changes from one packet to the next.*

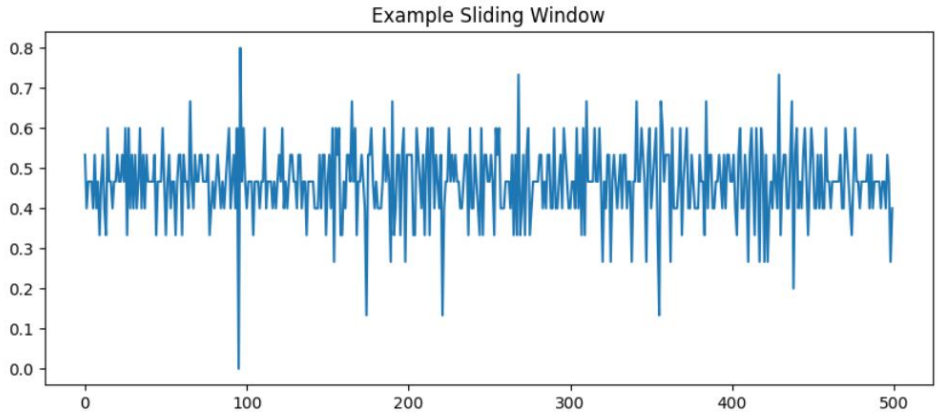



*Figure 9. Example sliding window used as model input after preprocessing. Each window becomes one training sample.*

The visual effect of preprocessing is consistent with the project brief: instead of forcing the classifier to memorize absolute signal levels, the pipeline encourages it to focus on fluctuation signatures generated by the deployment environment.

# 4. Model Training and Trusted Results

Two dedicated training notebooks were provided: one for CNN and one for ResNet. Both use the same-dataset evaluation strategy in which the window dataset is randomly split into train and validation partitions. These two notebooks produce the most trustworthy accuracy values in the current project because they actually load a dataset, fit a model, report metrics, and visualize training history.

## 4.1 CNN results

The CNN reached 82% validation accuracy, with weighted precision, recall, and F1-score all near 0.82. The class-wise report shows that bridge, forest, and lake are classified well, while garden is the most difficult class with an F1-score of 0.60. River also remains somewhat harder than bridge, forest, and lake. This pattern matches the exploratory plots: garden has fewer samples and appears statistically distinct but still variable, while river overlaps with other environments in absolute RSSI.

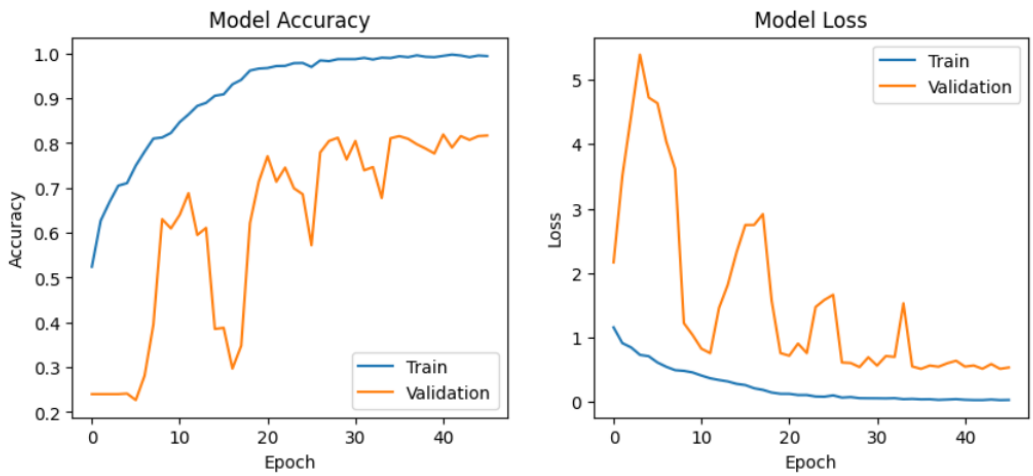



*Figure 10. CNN training curves. Training accuracy approaches 1.0 while validation accuracy stabilizes around 0.82, indicating useful learning but also some overfitting.*

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| bridge | 0.86 | 0.82 | 0.84 | 220 |
| forest | 0.86 | 0.88 | 0.87 | 198 |
| garden | 0.64 | 0.56 | 0.60 | 103 |
| lake | 0.90 | 0.86 | 0.88 | 168 |
| river | 0.72 | 0.86 | 0.78 | 136 |
| accuracy | | | 0.82 | 825 |
| macro avg | 0.80 | 0.80 | 0.79 | 825 |
| weighted avg | 0.82 | 0.82 | 0.82 | 825 |

*Figure 11. CNN classification report. Overall validation accuracy is 0.82, with garden as the most challenging class.*

A key observation from the CNN curves is the gap between training and validation performance. The model fits the training windows almost perfectly, but validation accuracy stabilizes much lower. This is not a failure, but it does show that the same-dataset split is easier than a real deployment setting and that generalization remains limited.

## 4.2 ResNet results

The ResNet notebook produced much stronger same-dataset results: 94% validation accuracy with macro and weighted F1-scores of 0.94. The class report indicates excellent performance across bridge, forest, lake, and river, while garden remains the weakest class at 0.88 F1. The confusion matrix confirms that most remaining errors are concentrated around garden, especially confusion with forest and lake.

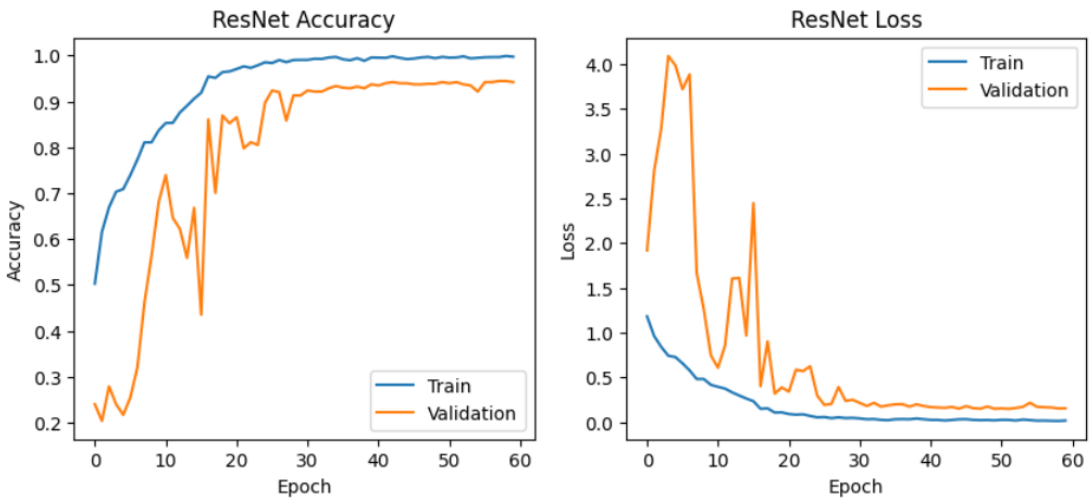
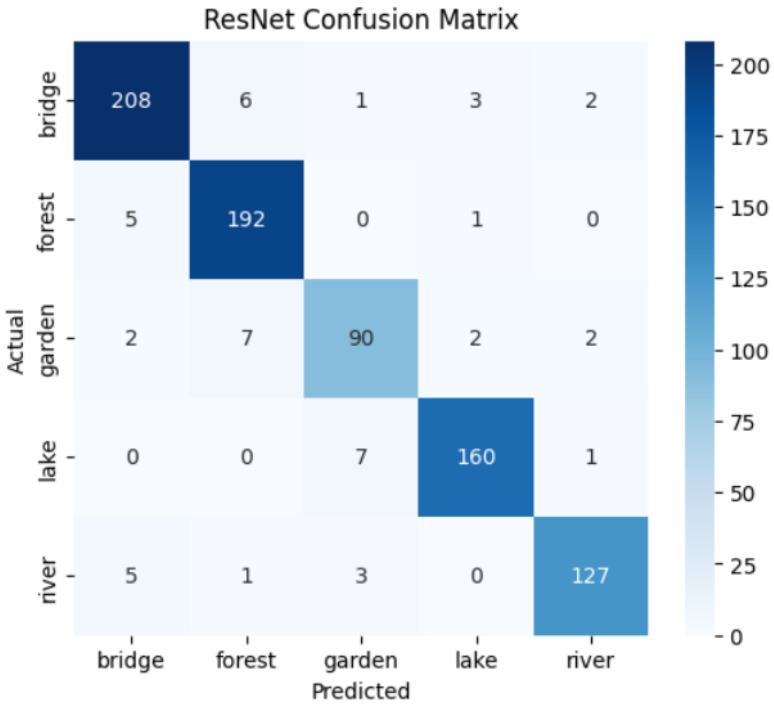



*Figure 12. ResNet training curves. Validation accuracy rises steadily and stabilizes near 0.94, while validation loss drops sharply after the early unstable epochs.*

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| bridge | 0.95 | 0.95 | 0.95 | 220 |
| forest | 0.93 | 0.97 | 0.95 | 198 |
| garden | 0.89 | 0.87 | 0.88 | 103 |
| lake | 0.96 | 0.95 | 0.96 | 168 |
| river | 0.96 | 0.93 | 0.95 | 136 |
| accuracy | | | 0.94 | 825 |
| macro avg | 0.94 | 0.94 | 0.94 | 825 |
| weighted avg | 0.94 | 0.94 | 0.94 | 825 |

*Figure 13. ResNet classification report. The model achieves 94% overall validation accuracy.*




```
19]: model.save("resnet_environment_model.h5")
```

*Figure 14. ResNet confusion matrix. The strongest confusion remains around garden, but most classes are predicted with high reliability.*

From the uploaded outputs alone, ResNet is the best performing same-dataset model in this project. However, this result should be reported carefully. Because the train/test split is random at the window level, overlapping windows from the same original run can appear in both train and test sets. This likely makes the validation task easier and can inflate measured accuracy. Therefore, 94% should be presented as strong within-dataset performance, not as guaranteed field generalization.

## 5. Feature-Importance Analysis

A Random Forest feature-importance analysis was used to inspect which positions inside the 500-sample RSSI window contribute most to classification. The full importance curve is relatively flat with many small peaks rather than one dominant spike. This is an important result: the environment signature is distributed across the entire temporal window, which supports the use of convolutional and residual models that aggregate local patterns across time.

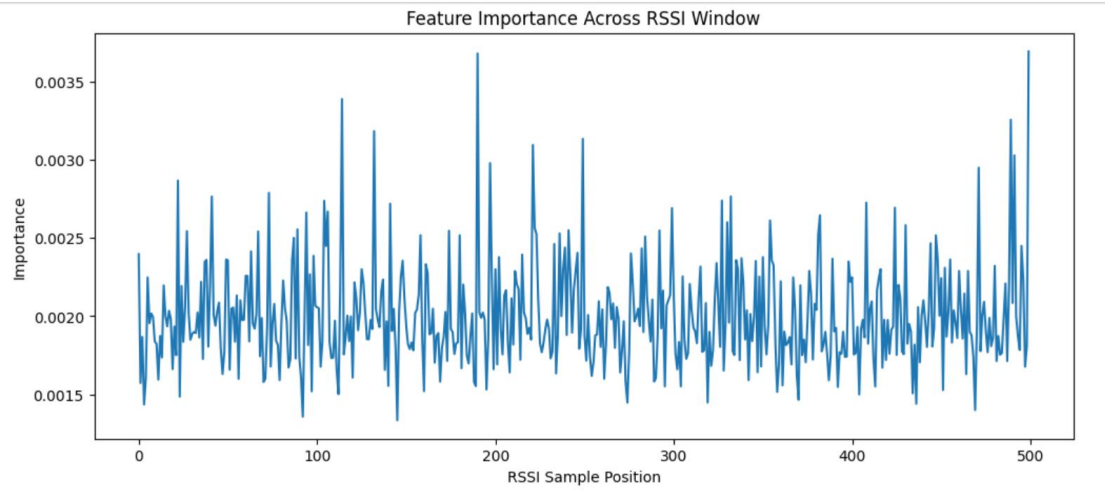



*Figure 15. Feature importance across the 500-sample RSSI window. Importance is spread across the sequence rather than concentrated at a single location.*

The top-ranked positions include samples near 22, 41, 73, 104, 114, 132, 190, 221, 249, 332, 489, 491, and 499. Notably, these positions appear throughout the window, including near both the beginning and the end. This again indicates that class information is global within the frame.

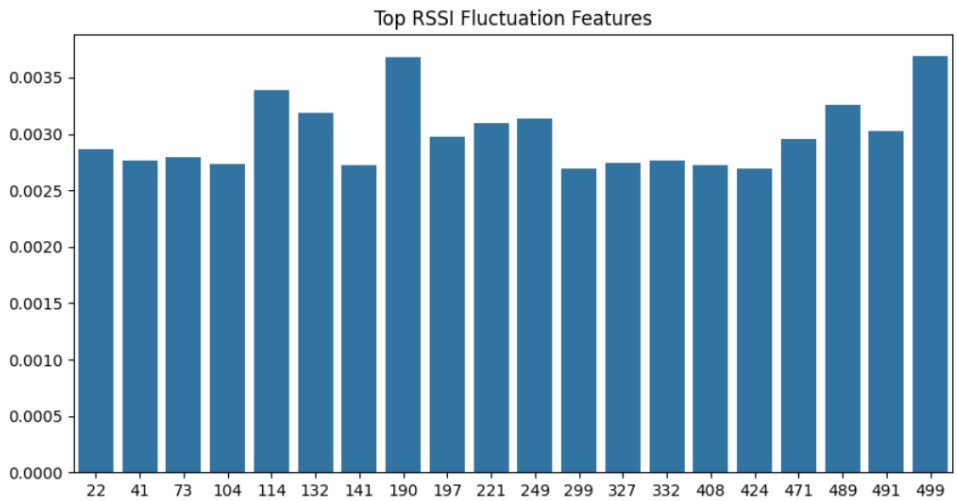



*Figure 16. Top twenty RSSI fluctuation positions ranked by Random Forest importance.*

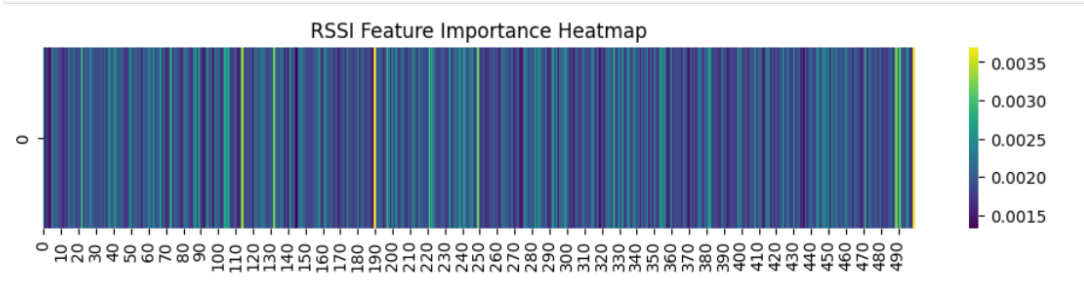



*Figure 17. Heatmap of feature importance across the full RSSI window.*

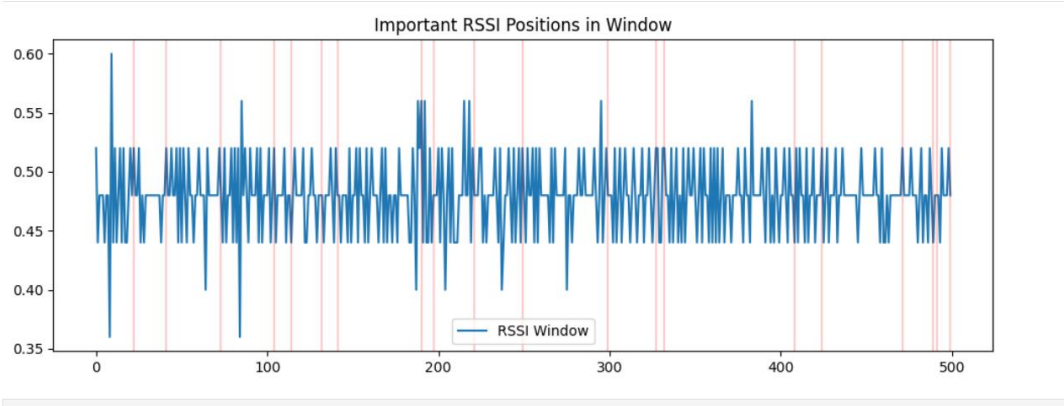



*Figure 18. Important RSSI positions overlaid on one example window. Discriminative positions are distributed throughout the sequence.*

This analysis is beneficial to the report because it provides interpretability. Instead of treating the deep-learning models as complete black boxes, it shows that meaningful temporal regions exist and that classification depends on repeated fluctuation structures across the window.

## 6. Frame-Length and Overlap Comparison: Current Status

A bar chart comparing CNN and ResNet across frame sizes was also uploaded. Superficially, it suggests that larger windows improve both models and that the difference between CNN and ResNet becomes smaller at 1000 samples. This trend is plausible and matches signal-processing intuition: longer windows capture more temporal context.

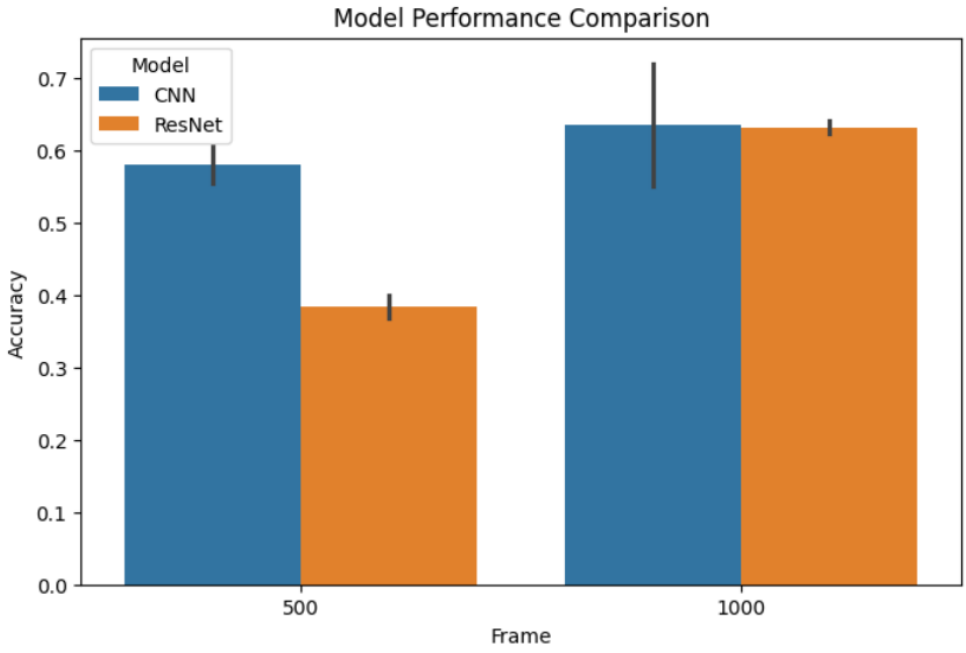



*Figure 19. Uploaded frame-size comparison chart. The trend suggests that 1000-sample windows are more informative than 500-sample windows.*

However, the underlying experiment notebook does not actually rebuild the dataset for each frame length and overlap value. Instead, it repeatedly loads one fixed file named windows_500_50.npz and then loops over nominal frame/overlap labels without regenerating the data. Therefore, the CSV and chart cannot yet be treated as final evidence for the required frame/overlap experiment. They are useful as exploratory outputs, but they should be marked as provisional until the experiment is rerun correctly.

## 7. Discussion

The CNN result of 82% shows that RSSI fluctuations are genuinely informative even with a comparatively simpler architecture. The ResNet result of 94% shows that richer temporal feature extraction can exploit those patterns very effectively. At the same time, the difference between 82% and 94% should not be overclaimed, because both values come from a same-dataset split at the window level. Until an unseen-environment split is run correctly, the project should avoid claiming that 94% will hold on a completely new deployment.

## 8. Opinion

RSSI fluctuations are not random noise only; they carry structured information about the surrounding environment. The strongest part of the work is the combination of preprocessing, deep-learning classification, and interpretability through feature-importance visualization.